Technical Report

Data Wrangling Report

Capstone Data - Supermarket Sales-Project

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1. Data Overview

Dataset size: 1,006 Rows × 16 Columns.

Content: Customer Invoices From Three Branches (A, B, C) With Attributes Such As Product Line, Unit Price, Quantity, Tax, Total, Date, Time, Payment Method, And Rating.

2. Data Cleaning Steps

Column Names:

Trimmed Leading/Trailing Spaces To Ensure Consistency.

Data Types:

Converted Unit Price And Quantity To Numeric.

Parsed Date Into Datetime Objects.

Extracted Hour, Year, Month, And Day For Time-Based Analysis.

Missing Values:

Dropped Rows With Missing Critical Values (Unit Price, Quantity, Or Date).

Imputed Missing Tax 5% And Total Using Calculated Formulas.

Created A New Column Total_Calc = Unit Price × Quantity For Consistency Checks.

For Missing Hour Values, Applied Random Imputation Within Business Hours (9 AM – 9 PM).

Feature Engineering:

Created Day_Period Categories (Night, Morning, Afternoon, Evening, Late Night).

Added New Time-Based Columns (Year, Month, Day).

3. Final Dataset

Final Shape:

1,001 Rows × 22 Columns (After Cleaning And Transformations).

Missing Values:

None Remaining In Key Columns (All Imputed Or Dropped).

Quality Check:

Dataset Is Now Consistent, Complete, And Ready For Exploratory Data Analysis (EDA).

4. Key Notes

Imputation Choice:

Random Imputation For Hour Ensured Completeness Without Biasing Toward A Single Value (E.G., Mode).

Consistency:

Both Total And Total_Calc Were Aligned To Validate Transaction Values.

Preparedness:

The Dataset Is Structured For Trend, Branch, Product, And Customer Analysis.